

Voice Agent OS

System Architecture

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A complete overview of the automated pipeline that transforms a client brief into production-ready voice agent configurations, powered by a self-improving knowledge brain.

Context

Inspra AI builds voice agents for businesses across multiple industries. Each engagement involves selecting the right platform, large language model, speech-to-text and text-to-speech providers, writing a production-grade system prompt, configuring the agent, running quality assurance checks, and delivering a complete package to the client. This process historically required significant manual effort and deep expertise across a growing ecosystem of tools.

The Voice Agent OS was created to automate and systematise this entire workflow. It acts as an intelligent operating system that ingests a client brief, runs research and build phases using specialised AI agents, validates the output through automated quality checks, and continuously learns from every engagement. The result is faster delivery, higher consistency, and a knowledge base that compounds with every project.

Executive Summary

The Voice Agent OS is a three-phase pipeline that takes a client brief as input and produces a fully configured, quality-assured voice agent as output. Phase 1 researches the optimal platform, model, and voice selections. Phase 2 writes the production system prompt using Inspra AI's proprietary prompt engineering methodology. Phase 3 validates every aspect of the build through automated QA, testing, and latency benchmarking. Underpinning the entire system is an Obsidian Wiki knowledge brain that stores learnings from every engagement and feeds them back into future builds. Four auto-learning loops ensure the system improves continuously without manual intervention.

How It Works: The Pipeline

Client Brief (Input). Every engagement starts with a client brief. This contains the business context, target audience, desired outcomes, and any platform preferences. The brief is the single input that triggers the entire pipeline.

Orchestrator (CLAUDE.md). The orchestrator ingests the brief, detects the industry and use case, and dispatches tasks to the appropriate phase agents. It reads from and writes to the wiki to ensure every decision is informed by prior learnings.

Phase 1: Research. Three agents run concurrently. The LLM selection agent evaluates models for speed, cost, and capability. The STT/TTS selection agent matches voice providers to the use case. The industry detection agent identifies sector-specific requirements.

Phase 2: Build. The prompt writer agent takes all research outputs and produces a production-grade system prompt using Inspra AI's inspra-voice methodology. This runs sequentially to ensure coherent, high-quality output.

Phase 3: Validate. Five agents run concurrently: QA checks for prompt quality, automated tests verify behaviour, diagram generation creates visual documentation, and latency benchmarking confirms the agent meets performance targets.

Client Output. The final deliverable includes production prompts, platform configuration, QA reports, and architecture diagrams. Everything the client needs to deploy.

Architecture Diagram

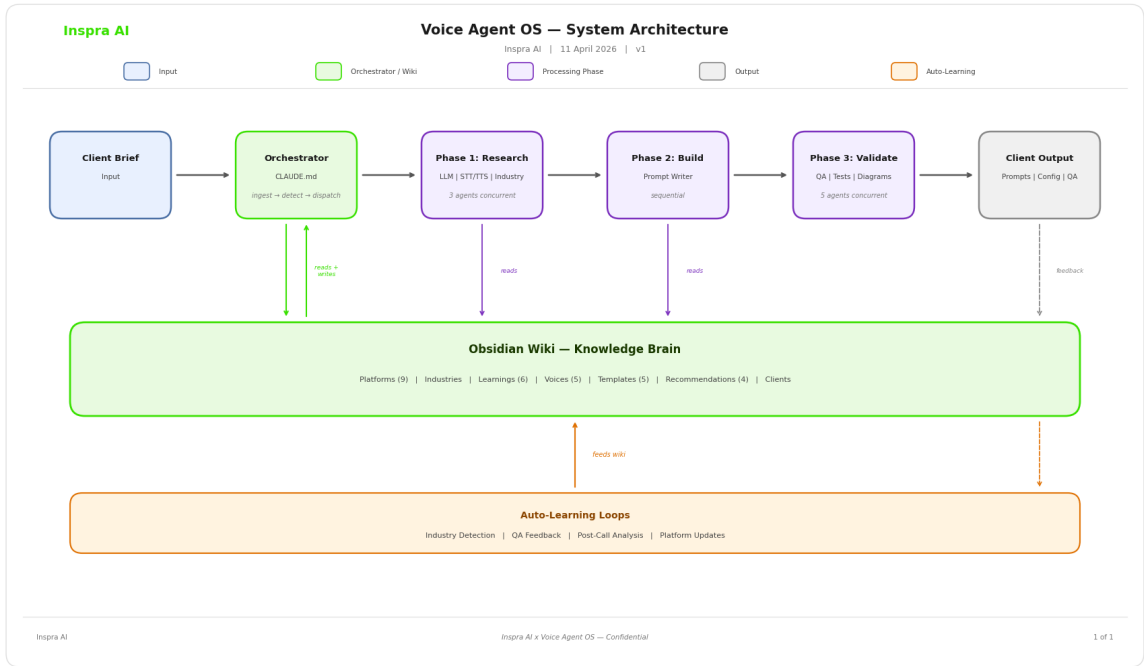


Figure 1: Voice Agent OS — System Architecture

The Wiki Brain

At the heart of the Voice Agent OS sits an Obsidian Wiki that functions as a living knowledge brain. It stores structured information across seven categories: platform configurations for nine supported platforms, industry-specific requirements, learnings captured from six core areas of expertise, voice profiles for five provider families, prompt templates for five common use cases, recommendation engines for four decision types, and client-specific histories.

The wiki is not a static reference. Every engagement reads from it and writes back to it. When the orchestrator processes a new brief, it consults the wiki for relevant precedents. When Phase 1 selects a platform or model, it checks the wiki's recommendation tables. When Phase 2 writes a prompt, it pulls from tested templates and learnings. After delivery, the output feeds back into the wiki, creating a compounding knowledge advantage that makes each subsequent engagement faster and more accurate.

Auto-Learning Loops

Industry Detection. Each new engagement refines the system's ability to identify industry-specific patterns and requirements automatically.

QA Feedback. Quality assurance findings from every build are catalogued and used to prevent recurring issues in future prompts.

Post-Call Analysis. Real-world call performance data is analysed to identify gaps between expected and actual agent behaviour.

Platform Updates. Changes to supported platforms (new features, API updates, pricing changes) are tracked and integrated into recommendation logic.

Summary Table

Component	Role	Impact
Client Brief	Input trigger for the pipeline	Standardised intake process
Orchestrator	Ingests, detects, dispatches tasks	Automated routing, zero manual triage
Phase 1: Research	LLM, STT/TTS, and industry selection	Optimal configuration in minutes
Phase 2: Build	Production prompt writing	Consistent, high-quality prompts
Phase 3: Validate	QA, testing, diagrams, latency	Defect-free delivery
Client Output	Final deliverable package	Complete, ready-to-deploy output
Wiki Brain	Knowledge storage and retrieval	Compounding expertise over time
Auto-Learning	Continuous improvement loops	System improves without manual effort

Next Steps

- Finalise the wiki knowledge base population for all nine supported platforms.
- Complete integration testing of the three-phase pipeline end to end.
- Onboard the first three client engagements through the automated pipeline.
- Establish post-call analysis data feeds from live deployments.
- Review and refine auto-learning loop accuracy after the first 20 engagements.